

RESEARCH PROPOSAL

**SSM-Adapter: A Bidirectional Selective State-Space
Temporal Adapter for Length-Robust Gloss-Free Sign
Language Translation**

A cross-backbone experimental programme on PHOENIX-2014T

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September 2026

Summary

Working title. SSM-Adapter: A Bidirectional Selective State-Space Temporal Adapter for Length-Robust Gloss-Free Sign Language Translation.

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The observation that motivates the study

Gloss-free sign language translation on PHOENIX-2014T shows a sharp drop in translation quality once the reference exceeds 12 words. On our PGAT-v1 reproduction, BLEU-4 falls from 17.39 (bin 7–12 words) to 5.03 (bin 13–18 words) — a 3.5x drop across one bin boundary. The pattern is documented across multiple published systems.

What we already ruled out

Suspect	Intervention tested	Outcome
Not enough visual evidence	Source32 (denser DINO features, wider prefix)	Retrieval improved; translation did not.
Prefix length too short	pgat-length v2 (variable-length prefix + mBART)	Flatter chrF, worse overall BLEU-4.
Decoder too small	Variable prefix + Qwen2.5-3B QLoRA	Mode collapse: two different videos → identical prediction.

The consistent failure signature across all three: fluent, domain-plausible German with weak dependence on the video.

What the study proposes

A small bidirectional selective state-space temporal adapter inserted at the visual-encoder output. Mamba-flavoured, pure PyTorch (no CUDA-kernel dependency), residual + LayerNorm. Ships as a drop-in module — the backbone architecture is unchanged; the training loss is unchanged.

What makes it defensible

The adapter is tested on two independent gloss-free SLT backbones under matched compute and matched scorers:

- pgat-length — mBART decoder + variable-length pose-guided prefix (ours).
- TSPNet — I3D features + fairseq transformer decoder (Li et al., NeurIPS 2020).

If the adapter reduces the length cliff on both, the effect is attributed to the adapter rather than to backbone-specific choices. If it reduces the cliff on one but not the other, the finding is reported honestly as backbone-specific.

Research questions

- RQ1. Does the SSM adapter reduce the length cliff on pgat-length?
- RQ2. Does the same adapter reproduce the effect on TSPNet?
- RQ3. Which hyperparameters matter — state dim, layer count, bidirectionality, residual scale?
- RQ4. How does the adapter compare against a parameter-matched transformer adapter at the same insertion point?

Progress at the time of writing

- Adapter implemented (src/pgat_length/models/ssm_adapter.py), smoke-tested at both d_model=512 (pgat-length) and d_model=1024 (TSPNet).
- pgat-length integration complete. DEV BLEU-4 7.61 (adapter) vs 7.36 (no adapter); chrF long-over-short ratio 0.807 vs 0.772 (flatter with adapter).
- TSPNet integration complete. Patched transformer_from_sign.py with backward-compatible --use-ssm-adapter flag. Baseline training in progress on the cluster; adapter variant queued.
- Cluster environment resolved (torch 1.11 + cu113, numpy pinned, sacrebleu 1.5.1, PyTorch floor-division patches, cython rebuild).

Three questions for the advisor

1. Is cross-backbone verification (two backbones, one dataset) sufficient scope for a master's thesis contribution, or is a second dataset (CSL-Daily) required for defense?
2. Should the primary claim be reported when the effect is present on BLEU-4 only, or when it is present on chrF and ROUGE-L (the current position, given the well-documented BLEU-4 lag on morphologically rich targets)?
3. Is the parameter-matched transformer control adapter a sufficient fairness baseline, or should additional controls (a randomly-initialised frozen adapter of matched parameters) be added?

Chapter 1 Introduction

1.1 Background of the Study

Sign language translation is a research problem at the intersection of computer vision, natural language processing, and accessibility. Its purpose is to translate signed videos into spoken-language text so that communication between Deaf or hard-of-hearing signers and hearing readers is supported (Camgoz et al., 2018, 2020). Gloss-free sign language translation is a stricter variant of the task: the model must translate directly from video to text without an intermediate gloss annotation. Removing gloss supervision is attractive because gloss annotation is expensive and language-specific, but it also removes a source of segment-level supervision that helped earlier gloss-based systems align visual content to output tokens (Zhou et al., 2023).

A specific failure mode of gloss-free translation on the PHOENIX-2014T benchmark is length sensitivity. When outputs are grouped by reference word count, translation quality is high on short sentences and drops sharply once the reference exceeds twelve words. On our own PGAT-v1 reproduction the BLEU-4 score for the seven-to-twelve word bin is 17.39 and the score for the thirteen-to-eighteen word bin is 5.03, a drop of more than three-fold across a single bin boundary. The pattern is not unique to a single system; Yazdani et al. (2026) report the same trajectory across several published gloss-free systems and Hamidullah et al. (2024) confirm it for their SEM-SLT model.

1.2 Motivation

The motivation for this study is both diagnostic and constructive. It is diagnostic because three separate architectural interventions on our local pipeline have failed to close the length cliff. Doubling visual evidence density (Source32) improved retrieval R@1 from 14.84 to 17.92 but did not improve translation quality. Introducing a variable-length visual prefix with a multilingual denoising decoder (pgat-length v2 with mBART) produced a flatter chrF curve at the cost of overall BLEU-4. Re-introducing a stronger causal decoder with parameter-efficient tuning (Qwen2.5-3B with QLoRA on the variable-length prefix) produced mode collapse: two distinct videos produced the identical output "dabei bleibt es meist trocken", and long-bin BLEU-4 fell below the mBART baseline. The common signature across the three failures is fluent, domain-plausible German that depends only weakly on the video.

The motivation is constructive because this signature suggests a specific missing ingredient. All three tested interventions changed only the visual capacity, the prefix length, or the decoder size while preserving the same short-range temporal encoder over the visual token sequence. State-space models offer a lightweight mechanism for long-range temporal integration with linear complexity in the sequence length, and Gu et al. (2022, 2023) have shown competitive performance against transformers on long-sequence video and audio understanding. The question this study asks is whether a small, backbone-agnostic state-space temporal adapter, inserted at the visual-encoder output without modifying the rest of the pipeline, can reduce the length cliff in gloss-free sign language translation.

1.3 Problem Statement

Gloss-free sign language translation on PHOENIX-2014T exhibits a reproducible length cliff on outputs longer than twelve words. This cliff has not been closed by a variety of architectural interventions targeting visual evidence, prefix length, or decoder size. Existing state-space models are proven for long-input classification and long-context language modelling, but they have not been applied to gloss-free SLT as a drop-in adapter across independent backbones under a controlled length-stratified evaluation.

The problem addressed in this study is that no controlled empirical evidence exists for whether a small bidirectional state-space temporal adapter can flatten the length curve of a gloss-free SLT system in a backbone-agnostic way. Without such evidence, it is not possible to conclude whether the limitation on long-sentence gloss-free translation is fundamentally a temporal-integration limitation or a broader semantic-composition limitation, and it is not possible to release a drop-in module that future gloss-free systems can adopt without redesign.

1.4 Research Gap

State-space models, cross-attention adapters, and temporal integration modules have each been studied in related work. The gap is the lack of focused investigation into their integration for gloss-free SLT with a specific evaluation on the length-cliff phenomenon and a specific requirement of backbone-agnostic transfer.

Two aspects of the gap are important for novelty positioning. First, existing state-space video work has largely reported classification metrics on datasets such as long-video benchmarks and dense video captioning, which measure different aspects of translation than the token-level fidelity a low-resource morphologically rich language such as German requires. Second, existing gloss-free SLT length-analysis work is descriptive rather than interventional: length sensitivity is measured across systems, but no single module is shown to reduce it consistently across systems. The proposed study fills both parts of the gap by treating the length cliff as an outcome to be reduced and by using two independent backbones so that the effect can be attributed to the adapter rather than to backbone characteristics.

1.5 Research Questions and Objectives

The main research question is: Can a small bidirectional selective state-space temporal adapter reduce the length cliff in gloss-free sign language translation in a backbone-agnostic manner?

The study is guided by four sub-research questions:

- RQ1: Does inserting a bidirectional selective state-space adapter at the visual-encoder output of the pgat-length pipeline (variable-length prefix + mBART) reduce the length cliff on PHOENIX-2014T?
- RQ2: Does inserting the same adapter into TSPNet's I3D-plus-transformer pipeline reproduce the reduction in length sensitivity?

- RQ3: Which adapter design choices matter most: state dimension, number of layers per direction, bidirectionality, or residual scaling?
- RQ4: How does the adapter compare against a parameter-matched transformer adapter placed at the same insertion point?

The main research objective is to design and evaluate SSM-Adapter, a lightweight bidirectional selective state-space temporal adapter for gloss-free sign language translation, and to validate its effect on the length cliff across two independent gloss-free SLT backbones.

The specific objectives are:

- To design and implement a Mamba-flavoured bidirectional selective state-space temporal adapter in pure PyTorch, with no CUDA-kernel dependency, that is safe to insert as a residual module after any visual-encoder that produces a (batch, time, channel) sequence.
- To integrate the adapter into two independent gloss-free SLT backbones, pgat-length and TSPNet, without modifying the backbone architecture or the training loss.
- To evaluate each backbone with and without the adapter under matched compute, matched hyperparameters, and matched scorers, reporting five-bin length-stratified BLEU-4, chrF, and ROUGE-L.
- To ablate state dimension, layer count, direction (bidirectional versus forward-only), residual scaling, and compare against a parameter-matched transformer adapter placed at the same insertion point.

1.6 Scope and Limitations

The study is scoped to gloss-free translation from sign video to spoken-language text on PHOENIX-2014T. The adapter is a small module inserted at the visual-encoder output; backbones are not modified beyond the insertion. The two backbones used for cross-backbone verification are pgat-length with a variable-length pose-guided prefix and an mBART decoder, and TSPNet with I3D features and a fairseq transformer decoder. Both use the same PHOENIX-2014T official train, validation, and test splits under the split-discipline convention that test is used only for the final frozen-architecture run.

The limitations are stated proactively. PHOENIX-2014T has limited signer diversity and a narrow weather-domain vocabulary; the study does not claim generalisation across sign languages or signing environments. Only two backbones are used for cross-backbone verification; a stronger claim of full backbone-agnostic behaviour would require additional systems, which is treated as future work rather than a project deliverable. The adapter's sequential state-space scan is implemented in pure PyTorch to avoid a CUDA-kernel build dependency, which is convenient but slower than the specialised Mamba kernel; wall time is not a claim of this thesis. Finally, the study focuses on translation-quality metrics; the adapter is not evaluated on inference latency or memory as primary criteria.

Chapter 2 SSM-Adapter Methodology

2.1 Research Approach

This study follows a design science and experimental research approach. It is design science because the main output is an artefact: SSM-Adapter, a bidirectional selective state-space temporal module for gloss-free sign language translation. It is experimental because the artefact is tested through controlled with-and-without comparisons on two independent backbones, ablations on the adapter's own hyperparameters, and a length-stratified evaluation designed to isolate the effect on the length cliff.

This approach is appropriate because the research questions cannot be answered through literature review alone. The thesis must implement the adapter, integrate it into two backbones, and measure whether it reduces the length cliff under matched compute and matched scorers. Any observed change must be attributable to the adapter rather than to backbone-specific choices, which is why the design uses two independent gloss-free SLT systems.

2.2 System Overview

SSM-Adapter is a single module with two integration points. In its first integration, the adapter is inserted into pgat-length, which combines a pose-guided variable-length prefix encoder with an mBART decoder. In its second integration, the adapter is inserted into TSPNet, which combines pre-extracted I3D features at three temporal scales with a fairseq transformer decoder. In both cases the insertion point is the same in principle: the adapter sits between the visual-encoder output and the downstream language decoder, applied to the (batch, time, channel) token sequence before the decoder attends to it.

2.3 Dataset and Input Representation

The study uses PHOENIX-2014T as the main benchmark. The input is a continuous German Sign Language video clip, and the target output is a German spoken-language sentence. The official train, validation, and test splits are used so that results remain comparable with prior work.

For each backbone, the visual representation is what the backbone already consumes:

- pgat-length: DINOv2 spatial features and TimeSformer motion features are extracted once and cached. A pose-guided tokenizer aggregates them into a variable-length prefix of length $K + 12$, where K depends on the source video length (typical $K \in [12, 32]$).
- TSPNet: pre-extracted I3D features at three temporal scales (span=8, span=12, span=16, all stride=2). At model time the three scales are projected to a common dimension, positional-embedded, and concatenated along the sequence axis.

2.4 Backbone Choice

Two backbones are used for cross-backbone verification. The choice is not arbitrary; it isolates the adapter effect from backbone characteristics that are otherwise easy to confuse with it.

pgat-length is our own implementation. It provides a modern denoising decoder (mBART-large-cc25) and a variable-length pose-guided prefix that already integrates spatial and motion information into a compact temporal sequence. Any adapter effect on top of pgat-length is measured against a competitive baseline in the range of BLEU-4 seven to eight on the PHOENIX-2014T test set.

TSPNet (Li et al., 2020) is an independent published gloss-free SLT system built on fairseq. It uses pre-extracted I3D features, a temporal semantic pyramid, and a transformer decoder trained from scratch. Its published test BLEU-4 is 12.98 under the local scorer, and its architecture is very different from pgat-length. If the adapter reduces the length cliff on both systems, the effect cannot be attributed to a specific backbone choice.

2.5 Adapter Design

The adapter is a bidirectional selective state-space module. Its input is a tensor of shape $[B, T, D]$ where B is the batch size, T is the sequence length, and D is the model hidden dimension. Its output is a tensor of the same shape. The design is intentionally minimal: the adapter has an input LayerNorm, a forward selective SSM block, a backward selective SSM block (applied to the time-reversed input and then flipped back), a summation of the two directions, dropout, a residual to the original tokens, and an output LayerNorm.

Each selective SSM block follows the Mamba structure (Gu & Dao, 2023) with three changes designed for portability. First, all operations use standard PyTorch primitives; there is no dependency on the specialised Mamba CUDA kernel. Second, the sequential scan is written as a Python loop over the time axis, which is negligible overhead at the small sequence lengths seen in gloss-free SLT ($T \leq 32$ for pgat-length, T up to a few hundred for TSPNet's concatenated three-scale sequence). Third, the state matrix A is initialised using the S4D-lin convention (Gu et al., 2022) for numerical stability with the input-dependent Δ .

2.5.1 Selective SSM block equations

Given an input $x \in \mathbb{R}^{(B \times T \times D)}$, one selective block computes:

$$[x_{ssm}, z] = W_{in}(x), \quad y = \sigma(DWConv_{\{d_{conv}\}}(x_{ssm}))$$

$$[\Delta, B, C] = W_{proj}(y), \quad \Delta = \text{softplus}(\Delta + \Delta_{bias})$$

$$\bar{A}_t = \exp(\Delta_t \cdot A), \quad \bar{B}_t = \Delta_t \cdot B_t$$

$$h_t = \bar{A}_t \odot h_{t-1} + \bar{B}_t \odot y_t, \quad o_t = C_t \cdot h_t$$

$$o = o + D \cdot y, \quad o = o \odot \sigma(z), \quad out = W_{out}(o)$$

The bidirectional wrapper computes o_fwd on the input, o_bwd on the time-reversed input flipped back to the original order, and returns $\text{LayerNorm}(x + s \cdot (o_fwd + o_bwd))$ where s is a residual scaling factor set to 1.0 by default. Padded time positions contribute zero to state updates because their inputs are already zero, and they are masked to zero in the output before the residual sum.

2.5.2 Default hyperparameters and parameter counts

Hyperparameter	Symbol	pgat-length	TSPNet
Model hidden dim	D	512	1024
State dim per channel	d_state	16	16
Depthwise conv kernel	d_conv	4	4
Inner-dim expansion	expand	2	2
Layers per direction	L	1	1
Dropout	p	0.1	0.1
Residual scale	s	1.0	1.0
Added trainable parameters	—	5.36 M	21.20 M

2.6 Integration into pgat-length

The adapter is inserted immediately after the pose-guided variable tokenizer produces temporal tokens $z_1, \dots, z_K$ of shape $[B, K, 512]$, with a segment-validity mask of shape $[B, K]$. The adapter transforms these tokens with its own residual and LayerNorm; the downstream articulator attention, global-summary attention, projection to mBART hidden dim, and mBART encoder-decoder pipeline are unchanged. The adapter's parameters are added to the pgat-side learning-rate group. When the adapter is disabled by configuration, the model reduces byte-for-byte to the pgat-length v2 baseline.

2.7 Integration into TSPNet

The adapter is inserted immediately after the three-scale visual features are concatenated along the sequence dimension and before fairseq's canonical (T, B, C) transpose. TSPNet's transformer encoder, cross-level attention mask, decoder, and evaluation harness are unchanged. The adapter is enabled via a single new command-line flag (`--use-ssm-adapter`); without the flag the encoder is byte-equivalent to the published TSPNet baseline. This keeps baseline reproduction cleanly isolated from adapter evaluation.

2.8 Training Procedure

For pgat-length, translation training uses the existing v2 configuration: warm start from the alignment stage's best checkpoint, mBART full fine-tune with variable prefix, label-smoothed cross-entropy at $\epsilon =$

0.15, AdamW with per-module learning rates, warmup ratio 0.10, cosine schedule, dropout 0.20 in the temporal encoder, and early stopping on development loss with patience six. When the adapter is enabled, its parameters share the pgat-side learning-rate group and the rest of the schedule is unchanged.

For TSPNet, training uses the authors' published run script: Adam with a fixed learning rate of $1e-4$, dropout 0.4, label smoothing 0.1, one encoder layer, reduce-on-plateau scheduler with patience eight, up to 200 epochs with early stopping, BLEU-selected best checkpoint. The adapter is enabled via the added command-line flag; the rest of the script is unchanged so that any observed change is attributable to the adapter.

2.9 Cross-Backbone Verification

The critical design element is that the same module is integrated into two backbones under matched local scorers. After training, each system's development predictions are scored with the pgat-length evaluation harness so that BLEU-4, chrF, ROUGE-L, and five-bin length metrics are computed identically across all four runs (pgat-length baseline, pgat-length with adapter, TSPNet baseline, TSPNet with adapter). If the adapter reduces the length cliff on both backbones, the finding is attributed to the adapter rather than to backbone-specific characteristics. If it reduces the length cliff on one but not the other, the finding is a backbone-specific effect and is reported honestly.

2.10 Methodological Limitations

The methodology is deliberately scoped. It uses only two backbones for cross-backbone verification; a stronger claim of full backbone-agnostic behaviour would require additional systems, which is left as future work. It uses one dataset (PHOENIX-2014T); confirmation on a second dataset such as CSL-Daily is desirable but not part of the primary deliverable due to compute and dataset access constraints. The adapter itself uses a pure-PyTorch sequential scan for portability; wall-time comparisons against the specialised Mamba CUDA kernel are not part of the claim. Finally, the study reports translation quality metrics as the primary outcome; interpretability of the SSM's learned state dynamics is not a claim of this thesis.

Chapter 3 Experimental Plan and Project Management

3.1 Baseline and Model Variants

The experimental design compares SSM-Adapter against controlled baselines on two independent backbones. The most important fairness requirement is that the parameter-matched transformer control variant must use the same total added parameter count as SSM-Adapter and be inserted at the same point in the pipeline. This prevents the experiment from confusing the benefit of state-space temporal integration with the benefit of simply adding more parameters at that location.

ID	Model variant	Purpose
E0	pgat-length v2 baseline (mBART, no adapter)	Same-backbone lower reference.
E1	pgat-length v2 with SSM-Adapter	Main pgat-length experimental system.
E2	pgat-length v2 with parameter-matched transformer adapter	Controls for added parameters vs SSM specifically.
E3	pgat-length v2 with forward-only SSM	Ablates bidirectionality.
E4	TSPNet baseline reproduction (I3D + transformer, no adapter)	Second-backbone lower reference.
E5	TSPNet with SSM-Adapter	Main TSPNet experimental system.
E6	TSPNet with parameter-matched transformer adapter	Controls for added parameters on the second backbone.

The primary claim is supported when E1 improves over E0 and E5 improves over E4 on the length-cliff dimension by an amount not explained by E2 relative to E0 or E6 relative to E4. The claim is falsified if either backbone shows no improvement or if the parameter-matched transformer control produces a comparable improvement.

3.2 Ablation Studies

Ablation	Comparison	Interpretation
Adapter presence	E1 vs E0, E5 vs E4	Primary effect of the module.
Parameter matching	E1 vs E2, E5 vs E6	Whether SSM structure matters beyond added parameters.
Bidirectionality	E1 vs E3	Whether both directions are needed.
State dim	$d_{\text{state}} \in \{8, 16, 32, 64\}$	Capacity-quality trade-off.
Layer count	$L \in \{1, 2, 3\}$ per direction	Depth-quality trade-off.
Residual scale	$s \in \{0.5, 1.0, 2.0\}$	Sensitivity to residual weighting.

Ablation	Comparison	Interpretation
Cross-backbone	Consistency of length-cliff reduction across pgat-length and TSPNet	Backbone-agnostic claim.

3.3 Metrics and Statistical Reporting

chrF is a primary metric because it captures character n-gram overlap and is suitable for German morphological variation. BLEU-1, BLEU-2, BLEU-3, and BLEU-4 are reported for comparability with prior SLT work, using SacreBLEU signatures for reproducibility. ROUGE-L is reported as a supporting sequence-overlap metric. All external baselines are rescored under the same scorer so cross-system comparisons are apples-to-apples.

The primary evaluation is a five-bin length-stratified analysis. Test samples are grouped by whitespace-tokenised reference word count into five bins (1-6, 7-12, 13-18, 19-24, 25-32 words) and each metric is reported per bin along with the overall corpus number. A single-number summary of the length cliff is the long-over-short ratio, defined as the mean metric on bins 13-32 divided by the mean metric on bins 1-12. A higher ratio indicates a flatter length curve.

All main results are reported as mean and standard deviation across three random seeds where compute allows. Statistical significance is assessed using paired bootstrap resampling with 2000 samples for the main comparisons: E1 versus E0, E1 versus E2, E5 versus E4, and E5 versus E6. Reported deltas include 95 percent bootstrap confidence intervals.

3.4 Feasibility and Compute Plan

The method is designed for a single 48 GB GPU (A6000). pgat-length training uses cached DINOv2 and TimeSformer features so no visual backbone runs during training; mBART fine-tune with a variable-length prefix and the SSM adapter fits in bf16 with gradient checkpointing at micro-batch three, gradient accumulation eight. TSPNet training uses pre-extracted I3D features and a small transformer decoder, easily fitting on the same GPU at max-tokens 4096.

Wall-time estimates per training run: on pgat-length, one full run is approximately five to eight hours until convergence via early stopping. On TSPNet, one full run is approximately four to eight hours to converge, with early stopping via reduce-on-plateau. The full experimental plan (E0 to E6 plus ablations across three seeds where compute allows) fits within roughly forty A6000-days.

3.5 Risk Management

The first major risk is that the adapter improves one backbone but not the other, weakening the cross-backbone claim. The mitigation is to report the negative result honestly and treat the finding as backbone-specific rather than backbone-agnostic; the release then documents the adapter as a candidate module and the paper narrative reduces its scope.

The second risk is TSPNet reproducibility. The published TSPNet code is a fairseq fork requiring old PyTorch and old NumPy pins. The mitigation is to install TSPNet in a separate conda environment with pinned versions (PyTorch 1.11, NumPy 1.19), apply small surgical patches for known PyTorch 1.11 API changes (integer floor division in the beam-search buffer), and use sacrebleu 1.5.1 to match the fairseq validation code. The successful local reproduction confirms this path is viable.

The third risk is that both backbones' improvements are small enough that the primary claim relies on chrF and ROUGE-L flattening rather than BLEU-4 improvement. The mitigation is to report all three metrics from the start, discuss the well-documented lag of BLEU-4 behind chrF on morphologically rich languages such as German, and commit to a specific plan for improving BLEU-4 through downstream techniques (length-penalty tuning, sequence-level knowledge distillation, or coverage penalties at inference).

3.6 Decision Rules and Contingency Plan

- Milestone 1: pgat-length baseline (E0) and pgat-length with SSM adapter (E1) both complete on the development split. If the length-cliff delta between E1 and E0 is not directional (flatter curve), the study pauses to inspect predictions for mode collapse before proceeding to the second backbone.
- Milestone 2: TSPNet baseline (E4) reproduces the published BLEU-4 near 12.98 under the local scorer within one BLEU-4 point. If it does not, the reproduction protocol is examined before running the adapter variant on TSPNet.
- Milestone 3: TSPNet with SSM adapter (E5) completes. If E5 improves length-cliff metrics over E4, the primary claim is supported; if it does not, the study is honestly reported as a backbone-specific improvement.
- Milestone 4: Parameter-matched transformer adapter controls (E2 and E6) complete. If E2 matches E1 and E6 matches E5, the SSM-specific claim is weakened and the discussion attributes the effect to the added-parameter position rather than the state-space dynamics.
- Milestone 5: Ablations complete on state dim, layers, direction, and residual scale. If no ablation shifts the length-cliff metric meaningfully, the adapter's effect is reported as robust to hyperparameter choice.
- Milestone 6: The final frozen-architecture PHOENIX-2014T test run is executed once on each system for the paper. Test scores are not used for design decisions.

3.7 Thesis Timeline

The revised thesis timeline runs from September 2026 to May 2027, with the SSM-Adapter experimental programme scheduled as the primary experimental contribution. Phase 1 (Sep-Oct 2026): baseline replication and pgat-length adapter runs. Phase 2 (Nov 2026): TSPNet reproduction and adapter runs. Phase 3 (Dec 2026-Feb 2027): ablations and cross-backbone analysis. Phase 4 (Mar-Apr 2027): writing. Phase 5 (May 2027): revision and submission.

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